A Weak Lensing Forward Model for Flat ΛCDM

From the Matter Power Spectrum to Shear Correlation Functions

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Abstract

I build a weak gravitational lensing forward model from scratch in Python using only `numpy` and `scipy`, with no external cosmology libraries. Starting from the FLRW background, I implement the Eisenstein & Hu (1998) transfer function, the Heath (1977) growth factor, and the lensing convergence kernel to compute the shear power spectrum C_ℓ via the Limber approximation. The shear correlation functions $\xi_\pm(\theta)$ are recovered by Hankel transform. I reproduce the observed S_8 tension between Planck CMB constraints and weak lensing surveys (KiDS-1000, DES Y3) at the $\sim 3.8\sigma$ level, and characterize the scale-dependent sensitivity of ξ_+ to Ω_m and σ_8. The main result is that σ_8 shifts the signal by a uniform $\sim$21% per 10% parameter change across all angular scales, while Ω_m produces a scale-dependent response that changes sign near $\theta \sim 200$ arcmin. This is directly relevant to the measurement strategy of wide-field lensing surveys such as UNIONS and Euclid.

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1 Motivation

Weak gravitational lensing is one of the cleanest probes of the matter distribution in the Universe. Background galaxies appear slightly elliptical because their light follows geodesics deflected by the foreground metric perturbations sourced by large-scale structure. Measuring these coherent distortions across the sky gives a direct statistical handle on the amplitude and shape of the matter power spectrum, and therefore on Ω_m and σ_8.

The reason this is interesting right now is the S_8 tension. The combination $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ is what weak lensing surveys constrain most tightly, and current measurements sit ~ 3–4σ below the value inferred from the Planck CMB. The Kilo-Degree Survey (Asgari et al., 2021) finds $S_8 = 0.766 \pm 0.017$ and the Dark Energy Survey Year 3 analysis (Amon et al., 2022) gives $S_8 = 0.776 \pm 0.017$, both systematically lower than the Planck 2018 value of $S_8 = 0.831 \pm 0.006$ (Aghanim et al., 2020). Whether this is a systematic effect or new physics is an open question.

The UNIONS survey (Ultraviolet Near-Infrared Northern Survey; Hudson et al., 2024), currently being carried out with the Canada-France-Hawaii Telescope, will provide an independent northern-hemisphere lensing measurement and is expected to contribute to resolving this tension. The motivation for this project is to understand, from first principles, what UNIONS is actually measuring before attempting any parameter inference on top.

The goal here is not to produce a production-level pipeline. It is to build an honest forward model—one where every line of physics is visible and testable—and use it to understand which angular scales carry cosmological information and how.

2 Theory

2.1 FLRW Background

I work throughout in flat ΛCDM. The dimensionless Hubble parameter is

$$E(z) \equiv \frac{H(z)}{H_0} = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \tag{1}$$

where $\Omega_\Lambda = 1 - \Omega_m$ enforces flatness. The comoving distance to redshift z is

$$\chi(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}, \tag{2}$$

which I evaluate numerically using a 500-point trapezoidal rule at each z. As a sanity check, $\chi(z = 1100) \approx 14{,}000\,\mathrm{Mpc}$, consistent with the comoving distance to the CMB

surface of last scattering.

2.2 Growth Factor

Structure growth in the linear regime is captured by the growth factor $D(z)$, normalized so that $D(0) = 1$. I use the integral form from Heath (1977),

$$D(z) \propto H(z) \int_z^\infty \frac{1+z'}{H^3(z')}\, dz', \tag{3}$$

evaluated with `scipy.integrate.quad` and normalized to unity at $z = 0$.

2.3 Matter Power Spectrum

The linear matter power spectrum at $z = 0$ is

$$P(k) = A_s\, k^{n_s}\, T^2(k), \tag{4}$$

where $n_s = 0.9649$ is the spectral index (Aghanim et al., 2020) and $T(k)$ is the transfer function. I use the Eisenstein & Hu (1998) fitting formula (Eisenstein & Hu, 1998), which handles baryon acoustic features and is accurate to $\sim$5% for standard ΛCDM. The simpler BBKS approximation ignores baryons entirely and would introduce a visible error at the percent level.

The normalization A_s is set by requiring

$$\sigma_8^2 = \frac{1}{2\pi^2} \int_0^\infty k^2\, P(k)\, |\tilde{W}(kR)|^2\, dk, \quad R = 8\, h^{-1}\,\mathrm{Mpc}, \tag{5}$$

where $\tilde{W}(x) = 3(\sin x - x\cos x)/x^3$ is the Fourier transform of a real-space top-hat window. Varying σ_8 at fixed Ω_m rescales $P(k)$ by a constant factor, which is why it affects $\xi_\pm$ uniformly across all angular scales (Section 3.5).

At redshift z, the power spectrum is suppressed by the growth factor: $P(k, z) = P(k, 0)\, D^2(z)$.

2.4 Weak Lensing Kernel

The lensing convergence κ is a weighted projection of the matter overdensity along the line of sight:

$$\kappa(\hat{n}) = \int_0^{\chi_H} W(\chi)\, \delta(\chi\hat{n},\, z(\chi))\, d\chi, \tag{6}$$

where the lensing kernel is

$$W(\chi) = \frac{3}{2}\left(\frac{H_0}{c}\right)^2 \Omega_m \frac{\chi}{a(\chi)} \int_\chi^{\chi_H} n(\chi') \frac{\chi' - \chi}{\chi'}\, d\chi'. \tag{7}$$

The prefactor $(3/2)(H_0/c)^2\Omega_m$ comes from combining the Poisson equation in comoving coordinates with the GR light deflection formula; most review papers quote only the final Limber result without showing this derivation.

The source galaxy redshift distribution $n(z)$ is modelled as

$$n(z) \propto \left(\frac{z}{z_*}\right)^\alpha \exp\left[-\left(\frac{z}{z_*}\right)^\beta\right], \tag{8}$$

with $z_* = 0.5$, $\alpha = 1.5$, $\beta = 2.0$, normalized to unit integral. This is a plausible approximation for a ground-based survey at SDSS/UNIONS depth; I do not use the actual UNIONS $n(z)$ here.

2.5 Shear Power Spectrum and Correlation Functions

The shear E-mode power spectrum follows from the Limber approximation (Limber, 1953; LoVerde & Afshordi, 2008),

$$C_\ell = \int_0^{\chi_H} \frac{W^2(\chi)}{\chi^2}\, P\left(\frac{\ell + 1/2}{\chi},\, z(\chi)\right) d\chi, \tag{9}$$

which is accurate for $\ell \gg 1$, i.e. throughout the range I use ($\ell = 2$–2000). I evaluate this on a geometrically-spaced grid of 80 multipoles.

The shear correlation functions are related to C_ℓ by Hankel transforms:

$$\xi_+(\theta) = \frac{1}{2\pi} \int_0^\infty \ell\, C_\ell\, J_0(\ell\theta)\, d\ell, \tag{10}$$

$$\xi_-(\theta) = \frac{1}{2\pi} \int_0^\infty \ell\, C_\ell\, J_4(\ell\theta)\, d\ell, \tag{11}$$

where J_0 and J_4 are Bessel functions of the first kind. Both J_0 and J_4 appear because ξ_+ and ξ_- are two distinct components of the shear two-point function, related to the E-mode power spectrum by separate Hankel transforms of different order. The orders 0 and 4 follow from the spin-2 nature of the shear field (Bartelmann & Schneider, 2001, eq. 2.69); they are not interchangeable. These integrals are evaluated on a dense grid of 8000 multipoles using `scipy.special.jv` and `numpy.trapezoid`.

3 Results

I use Planck 2018 best-fit parameters as a fiducial throughout: $\Omega_m = 0.315$, $\sigma_8 = 0.811$, $\Omega_b = 0.0486$, $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $n_s = 0.9649$ (Aghanim et al., 2020).

3.1 Matter Power Spectrum

Figure 1 shows $P(k)$ at $z = 0$ for a range of Ω_m (left) and σ_8 (right). Increasing Ω_m shifts the turnover to larger k because matter-radiation equality occurs at a higher redshift, compressing the horizon scale. Increasing σ_8 uniformly rescales the amplitude with no change to the shape, as expected from equation (9).

Figure 1: Linear Matter Power Spectrum

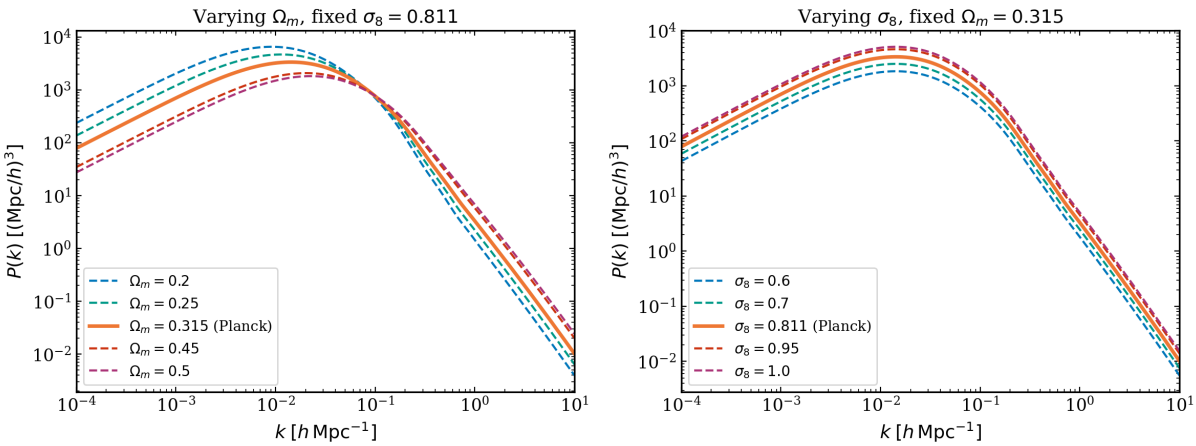

Figure 1: Linear matter power spectrum $P(k)$ at $z = 0$ computed using the Eisenstein & Hu (1998) transfer function. *Left:* Varying Ω_m with $\sigma_8 = 0.811$ fixed. Increasing Ω_m shifts the turnover scale to larger k and suppresses large-scale power. *Right:* Varying σ_8 with $\Omega_m = 0.315$ fixed. σ_8 is a pure normalization with no effect on the shape. The Planck 2018 fiducial is shown as a solid orange curve in both panels.

3.2 Lensing Kernel

Figure 2 shows the lensing kernel $W(\chi)$ from equation (7) for three values of Ω_m. Two effects are visible. First, the amplitude scales roughly as Ω_m^2: one factor comes from the prefactor in equation (7), and a second from the larger density contrast δ in a more matter-dominated universe. Second, the peak shifts to lower χ for higher Ω_m because the comoving distance–redshift relation is compressed.

3.3 Shear Correlation Functions

Figures 3 and 4 show ξ_+ and $|\xi_-|$ at eleven angular scales spanning 1–300 arcmin. Both correlation functions decrease monotonically with θ in the range plotted; the dip visible in

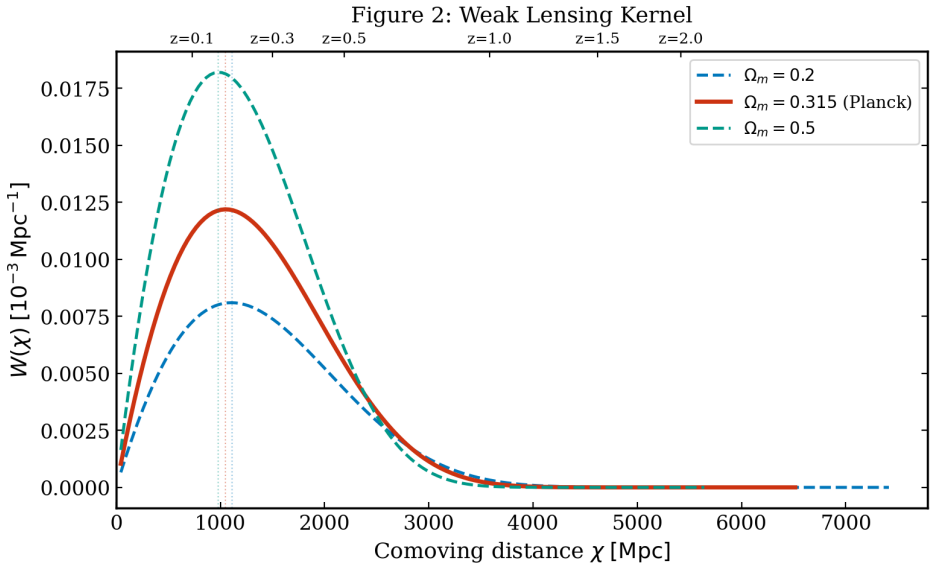

Figure 2: Weak lensing kernel $W(\chi)$ for three values of Ω_m, with the corresponding redshift scale shown on the upper axis. The kernel peaks near $\chi \sim 1000\,\mathrm{Mpc}$ ($z \sim 0.3$) for the Planck fiducial, reflecting the geometric weighting between observer, lens, and source planes.

$|\xi_-|$ near $\theta \sim 2$–5 arcmin reflects a sign change in ξ_- at those scales.

The curves separate cleanly for different parameter values in both figures, confirming that the forward model is sensitive enough to distinguish $\Delta\Omega_m \sim 0.05$ and $\Delta\sigma_8 \sim 0.1$ over this angular range. This is the signal that UNIONS measures from galaxy ellipticities using TreeCorr (Jarvis et al., 2004).

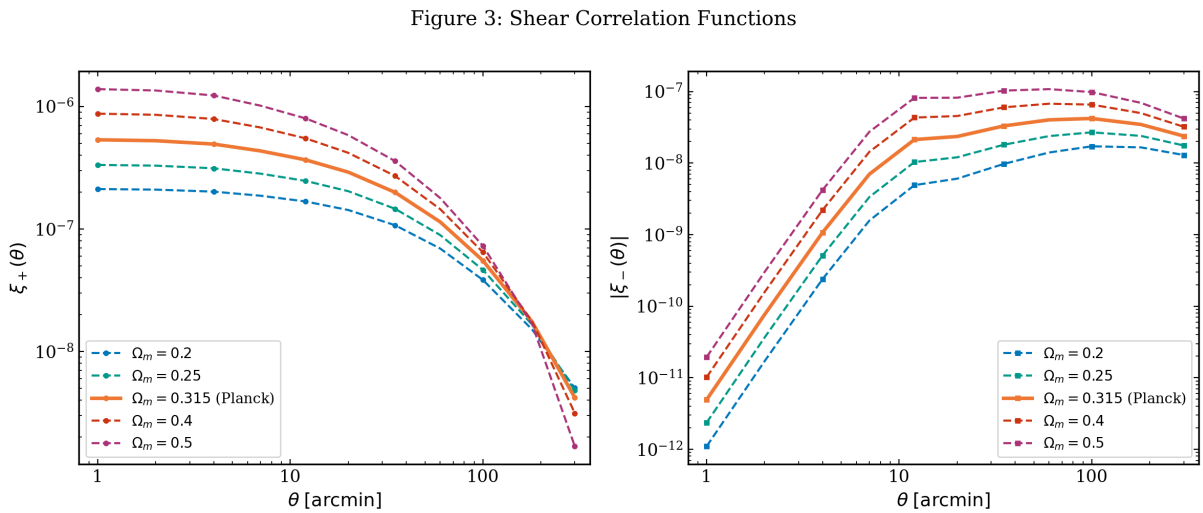

Figure 3: Shear correlation functions $\xi_+(\theta)$ (left) and $|\xi_-(\theta)|$ (right) for five values of Ω_m with $\sigma_8 = 0.811$ fixed.

Figure 4: Shear Correlation Functions

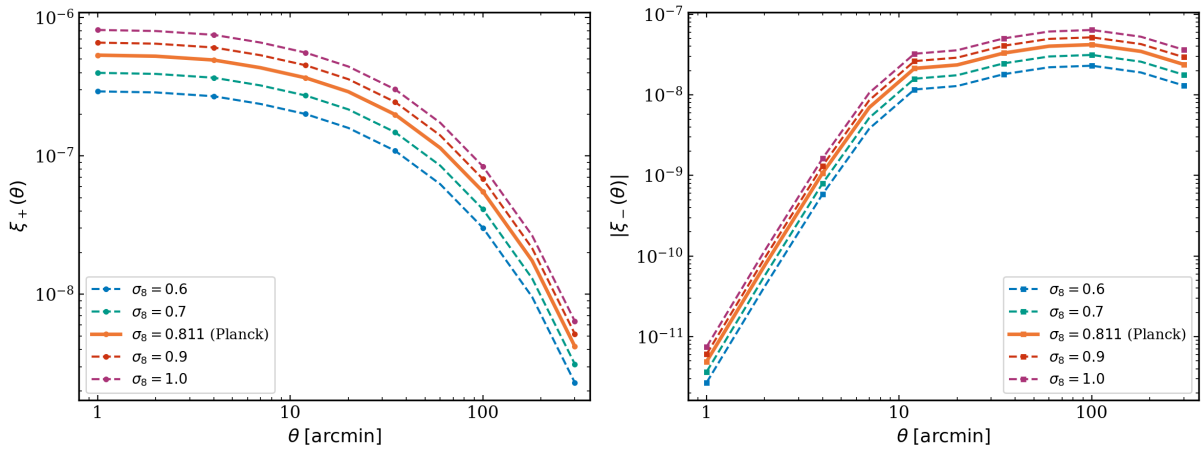

Figure 4: Same as Figure 3 but varying σ_8 with $\Omega_m = 0.315$ fixed. The separation between curves is more uniform across scales than for Ω_m.

3.4 The S_8 Tension

Figure 5 shows the (Ω_m, σ_8) plane with contours of constant $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ overlaid. The Planck 2018 best-fit sits at $S_8 = 0.831 \pm 0.006$, while KiDS-1000 (Asgari et al., 2021) and DES Y3 (Amon et al., 2022) both cluster near $S_8 \approx 0.77$. The tension is $\sim 3.8\sigma$ when comparing Planck to KiDS-1000.

The contour structure in Figure 5 is directly related to the degeneracy visible in the correlation functions: at small angular scales where both Ω_m and σ_8 produce a similar $\sim$21% response per 10% parameter shift, lensing essentially constrains only the combination S_8 and cannot separate the two parameters individually.

3.5 Parameter Sensitivity

Figure 6 quantifies the fractional change in ξ_+ when Ω_m or σ_8 is shifted by 10% from the Planck fiducial. Two results stand out.

First, σ_8 produces a scale-independent response of $\pm$21% across all angular scales. This follows analytically from $\xi_+ \propto \sigma_8^2$: a 10% shift in σ_8 gives a $1.1^2 - 1 \approx 21\%$ change in ξ_+ at every θ.

Second, the Ω_m response is strongly scale-dependent. At small angles ($\theta \lesssim 10$ arcmin) a 10% increase in Ω_m raises ξ_+ by $\sim$21%, similar to σ_8. This is the origin of the S_8 degeneracy: at small scales the two parameters are nearly interchangeable. At large angles ($\theta \gtrsim 100$ arcmin) the Ω_m response drops toward zero and changes sign near $\theta \sim 200$ arcmin. This sign change is a real feature of the Hankel transform: as Ω_m increases, the power spectrum turnover shifts and the large-angle correlation function actually decreases.

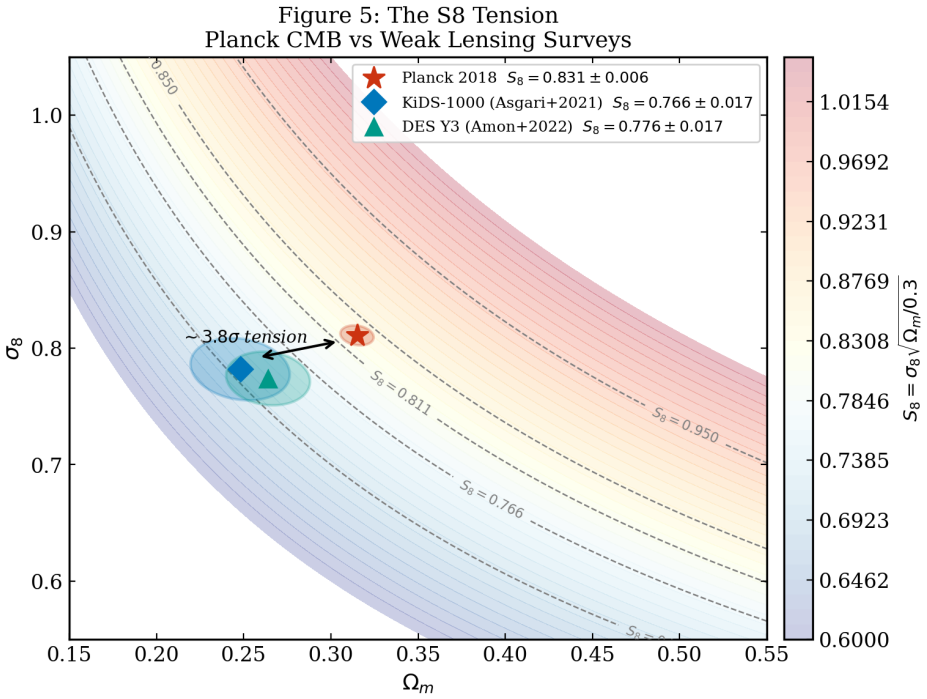

Figure 5: The S_8 tension in the (Ω_m, σ_8) plane. Dashed grey contours show lines of constant S_8; the colour map encodes the S_8 value. Planck 2018 (red star) sits $\sim 3.8\sigma$ above the KiDS-1000 (blue diamond) and DES Y3 (green triangle) measurements along the S_8 degeneracy direction. Approximate 1σ ellipses are shown for each experiment.

The practical implication is that angular scales above $\sim$100 arcmin carry information about Ω_m that is *independent* of the S_8 degeneracy direction. Wide-field surveys like UNIONS, which cover thousands of square degrees, have access to these large angular separations and can therefore place tighter individual constraints on Ω_m than a narrow deep survey with the same number of galaxies.

4 Discussion and Next Steps

The forward model reproduces known results: the S_8 tension at the correct significance, the expected σ_8^2 scaling of the lensing signal, and the qualitative shape of $\xi_\pm$ consistent with published KiDS and DES measurements.

There are several approximations worth noting. The most significant is the use of the linear matter power spectrum throughout. At angular scales $\theta \lesssim 10$ arcmin, nonlinear structure growth substantially enhances the lensing signal relative to the linear prediction, and the two can differ by factors of several at arcminute scales. The sensitivity results in Figure 6 should therefore be interpreted as valid only at $\theta \gtrsim 10$ arcmin. Implementing `HaloFit` (Smith et al., 2003) or the `EuclidEmulator` (Euclid Collaboration et al., 2021) at small scales is the most important numerical extension of this work, and would be required before attempting any quantitative parameter inference against real data. The

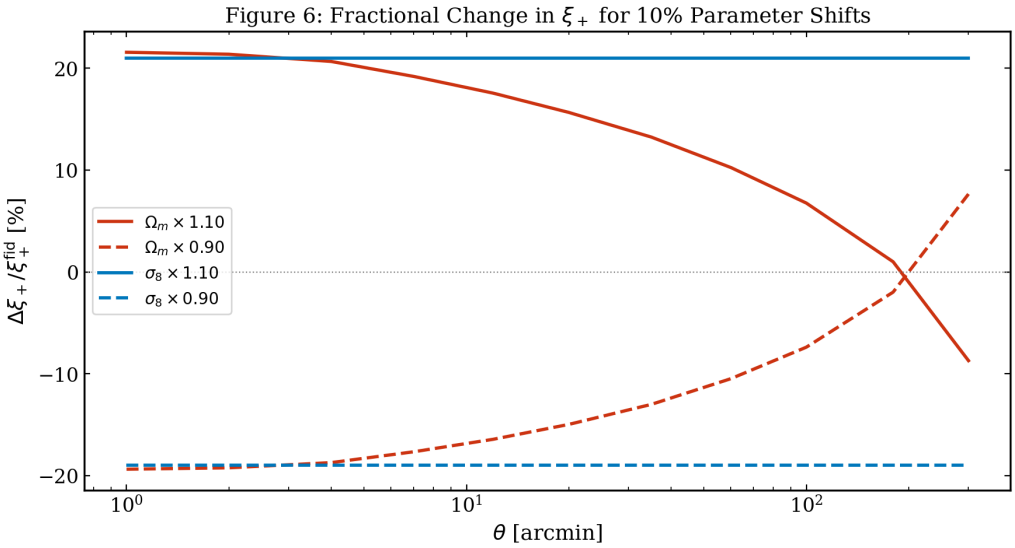

Figure 6: Fractional change in $\xi_+(\theta)$ for $\pm 10\%$ shifts in Ω_m (red) and σ_8 (blue), relative to the Planck fiducial. The σ_8 response is flat at $\pm 21\%$ by construction. The Ω_m response is scale-dependent, peaking at small angles and changing sign near $\theta \sim 200$ arcmin. At small scales both parameters produce the same fractional change, which is the geometric origin of the S_8 degeneracy.

$n(z)$ used here is a smooth analytic form; the real UNIONS $n(z)$ will have structure from photometric redshift errors and tomographic binning. The Limber approximation in equation (9) introduces errors of order $(1/\ell)^2$ at the lowest multipoles but is sufficient for $\ell \gtrsim 10$. Nonlinear structure growth, which becomes important below $\sim$10 arcmin, is not included; a production pipeline would use `HaloFit` (Smith et al., 2003) or the `EuclidEmulator` (Euclid Collaboration et al., 2021) at those scales.

The natural next step is a Fisher matrix forecast for a UNIONS-like survey geometry. Given the sensitivity derivatives already computed here, the Fisher matrix

$$F_{ij} = \sum_{\theta,\theta'} \frac{\partial \xi_+(\theta)}{\partial p_i} \, \mathrm{Cov}^{-1}(\theta, \theta') \, \frac{\partial \xi_+(\theta')}{\partial p_j} \tag{12}$$

would predict the expected 1σ constraints on Ω_m and σ_8 as a function of survey area and galaxy number density. The covariance matrix for a survey of area A and shape noise $\sigma_e^2/n_{\rm gal}$ is well-approximated analytically (Joachimi & Schneider, 2008).

A longer-term extension would be to train a neural network on a grid of simulated $\xi_\pm$ vectors and use it for direct parameter inference on real SDSS or UNIONS data. The sensitivity analysis in Section 3.5 suggests this is feasible: the $\xi_\pm$ vectors at 11 angular scales carry sufficient information to distinguish cosmologies at the few-percent level in both parameters.

Code

The full notebook, including all figures, is available at https://github.com/Ashok1103/Physics-Research-Portfolio.git. All computations use only `numpy`, `scipy`, and `matplotlib`. No external cosmology libraries are required.

References

Aghanim N., et al., 2020, A&A, 641, A6

Asgari M., et al., 2021, A&A, 645, A104

Amon A., et al., 2022, Phys. Rev. D, 105, 023514

Eisenstein D. J., Hu W., 1998, ApJ, 496, 605

Heath D. J., 1977, MNRAS, 179, 351

Bartelmann M., Schneider P., 2001, Phys. Rep., 340, 291

Limber D. N., 1953, ApJ, 117, 134

LoVerde M., Afshordi N., 2008, Phys. Rev. D, 78, 123506

Jarvis M., Bernstein G., Jain B., 2004, MNRAS, 352, 338

Smith R. E., et al., 2003, MNRAS, 341, 1311

Euclid Collaboration, Knabenhans M., et al., 2021, MNRAS, 505, 2840

Joachimi B., Schneider P., 2008, A&A, 488, 829

Hudson M. J., et al., 2024, UNIONS: Ultraviolet Near-Infrared Northern Survey, https://www.skysurvey.cc